

Digital Signal Processing: Lab 2 Report

Joshua Claassen (jec1g24)

April 2026

1

The rectangular (boxcar) window produced the thickest (least sharp) and least smooth chirp line. Although, in theory, the rectangular window has the narrowest main lobe in the frequency domain, its high sidelobes (caused by very low sidelobe suppression) introduces significant spectral leakage making the chirp line appear blurry and wider.

The hamming window produced a slightly sharper and smoother chirp line, indicating reduced spectral leakage and better sidelobe suppression compared to boxcar.

The hann and blackman windows produced the cleanest and sharpest chirp lines, with very little spectral spreading due to high sidelobe suppression. They appeared nearly identical at this FFT resolution.

Window length:

Longer windows mean the FFT has finer frequency bins, making it easier to distinguish closer frequencies, thus having a better frequency resolution. But fast changes in the signal may get lost since you're sampling over a longer time, so a higher frequency resolution comes at a trade-off for a worse time resolution.

On the other hand, shorter windows lead to wider FFT frequency bins. This means frequency estimates becomes less precise (worse frequency resolution), but you can tell more accurately when changes in the signal occur (better time resolution).

On the chirp spectrogram, frequency resolution is reflected in the vertical thickness of the chirp line, while time resolution is reflected in the horizontal sharpness of the line.

Window shape:

The window shape mainly affects frequency resolution by spectral leakage. A rectangular (boxcar) window has a narrow main lobe (good theoretical frequency resolution) but very high sidelobes, so energy spreads widely into nearby frequencies. This leakage can actually make frequency estimates look less clean in practice, as shown by the graph. Tapered windows (like Hann, Hamming and Blackman) reduce sidelobes, so there is less leakage and peaks are cleaner, but their main lobes are wider, which slightly reduces the ability to distinguish very close frequencies.

In the STFT, each time frame is multiplied by a window, and tapered windows (like Hann, Hamming, and Blackman) reduce the contribution of samples near the edges of each segment. This effectively means each time slice is weighted more toward the centre of the window, which can make transitions between frames look slightly smoother.

Rectangular windows preserve abrupt changes most directly (best time resolution), but can look noisier due to leakage. Tapered windows slightly blend adjacent time samples because of edge weighting, which can make rapid changes appear a bit less sharp in time (slightly worse time resolution).

2

$\gamma = 0.8$ compressed the spectral dynamic range, whereas $\gamma = 1.2$ expanded it. $\gamma = 0.8$ resulted in a warmer and fuller audio, with notes being more connected. This is consistent with the resulting spectrogram, which looked more filled in since weaker background energy was being emphasised.

In contrast, $\gamma = 0.8$ resulted in a brighter audio with notes being more separated. This could be seen in the spectrogram which looked cleaner and sparser. It emphasised dominant harmonics and low-level background energy seemed to disappear.

RMS normalisation ensured that volume stayed roughly constant, so the spectral shape could be fairly contrasted.

3

Amplitude modulation only affects the amplitude of the carrier wave, so its frequency remains the same (5000 Hz). Though, the nature of multiplying the carrier by the message does cause additional frequency components in the AM spectrum (carrier freq. $\pm$ message freq.).

Frequency modulation on the other hand, doesn't affect the amplitude, only the frequency, of the carrier wave itself, causing it to speed up or slow down. As a result energy is spread across many more frequency components, making the FM spectrum wider than the AM spectrum despite the same message frequency of 200 Hz.

4

Voiced phonemes appear in regions of high energy and low ZCR since these sounds use the vocal folds which produce smooth and regular waveforms with fewer sign changes. Unvoiced phonemes appear in higher ZCR regions and have lower and less structured energy because these sounds are generated by turbulent airflow and not vocal fold vibration, resulting in frequent sign changes.

5

As σ_x decreases, Δk increases. $\Delta x \Delta k = 0.5$ to 4 significant figures, demonstrating an inverse relationship between Δx and Δk .

$$p = \hbar k$$

$$\Delta p = \hbar \Delta k$$

Since $\hbar = 1$:

$$\Delta p = \Delta k$$

$$1 : \Delta x \Delta p = \Delta x \Delta k$$

$$\Delta x \Delta p \geq \frac{\hbar}{2}$$

With $\hbar = 1$:

$$\Delta x \Delta p \geq \frac{1}{2}$$

By 1:

$$\Delta x \Delta k \geq \frac{1}{2}$$

True since $\Delta x \Delta k = \frac{1}{2}$, so $\Delta x \Delta p \geq \frac{\hbar}{2}$ holds.

Heisenberg's uncertainty principle states that there is a fundamental limit to how precisely position and momentum can be known at the same time, arising from the wave nature of quantum systems. This relationship shows that increasing the certainty in a particle's momentum must inversely reduce the certainty in its position and vice versa.